

1- Which one of the following increases the volume of the thoracic cage?

- A) Relaxation of the diaphragm
- B) Contraction of internal intercostal muscles
- ✓ ☒ C) Contraction of the sternocleidomastoid muscles
- D) Relaxation of the scalenus muscles

2- What is the normal alveolar pressure (cmH₂O) at the beginning and end of normal inspiration, respectively?

- A) 0 and -1
- B) -1 and 0
- ✓ ☒ C) 0 and 0
- D) -1 and -1

3- Which one of the following leads to increase in diameter of airways?

- A) histamine
- ✓ ☒ B) Activation of β_2 adrenergic receptors
- C) Stimulation of M3 receptors
- D) Vagu reflex

4- If the tidal volume is 600 ml for a change in transpulmonary pressure of 3 cmH₂O, how much will the lung compliance be?

- A) 100 ml cmH₂O
- ✓ ☒ B) 200 ml/cmH₂O
- C) 800 ml/cmH₂O
- D) 1800 ml/cmH₂O

5- What is the "Expiratory Reserve Volume"?

- A) The volume of air remaining in the lungs after the most forceful expiration
- B) The maximum amount of air a person can expel from the lungs after first filling the lungs to their maximum extent
- C) The amount of air that remains in the lungs at the end of normal expiration
- ✓ ☒ D) The extra volume of air that can be expired by deep exhalation

6- Which zone of the lung has continued blood flow?

- A) Zone-I at the top of the upright lung
- B) Zone-II at the top of the upright lung
- ✓ ☒ C) Zone-III at the bottom of the upright lung
- D) Zone-II during exercise

7- In normal condition, what are the alveolar and arterial blood O₂ pressures, respectively?

- A) 104 and 104 mmHg
- B) 95 and 95 mmHg
- C) 95 and 104 mmHg
- ✓ ☒ D) 104 and 95 mmHg

8- What is the hypoxia-induced vasoconstriction mechanism?

- A) A decrease in alveolar oxygen pressure to less than 60 mm Hg causes increased blood flow in the pulmonary capillaries.
- ✓ ☒ B) A decrease in alveolar oxygen pressure to less than 73 mm Hg causes decreased blood flow in the pulmonary capillaries.
- C) A decrease in alveolar oxygen pressure to less than 104 mm Hg causes decreased blood flow in the pulmonary capillaries.
- D) A decrease in alveolar oxygen pressure to less than 73 mm Hg causes increased blood flow in the pulmonary capillaries.

9- What happens to the rate of O₂ diffusion when the respiratory membrane surface is reduced by half?

- A) Doubles
- B) Decreases to one sixteenth (1/16)
- C) Increases 16-fold
- ✓ ☒ D) Decreases by half

10- Which one of the following centers can generate the ramp signal?

- ✓ ☒ A) Dorsal respiratory group (DRG)
- B) Ventral respiratory group (VRG)
- C) Pneumotaxic center
- D) Apneustic center

11- Which one of the following is likely the primary stimulus of the neurons in chemosensitive area?

- ✓ ☒ A) Oxygen
- ☒ B) Hydrogen ions
- C) Carbon monoxide
- D) Nitrogen

12- Which ion has a higher concentration in RBCs of venous blood than arterial blood?

- ✓ ☒ A) Cl^-
- B) HCO_3^-
- C) H^+
- D) Fe^{2+}

13- What condition causes a physiologic shunt?

- A) A decrease in ventilation/perfusion ratio (V_A/Q) to 0.6 at the top of the upright lung
- B) Mixture of pulmonary vein blood with bronchial vein blood
- ✓ ☒ C) An increase in V_A/Q ratio to 2.5 at the bottom of the upright lung
- D) A decrease in ventilation/perfusion ratio (V_A/Q) to 0.6 at the bottom of the upright lung

14- Which one the following is correct about the oxygen utilization coefficient?

- A) It increases only to twice during heavy exercise.
- ✓ ☒ B) It can triple during heavy exercise.
- C) At rest, its normal level is 40 %.
- D) With an increase in the body's metabolism, the amount of it decreases.

15- What factor increases lung compliance?

- A) An increase in the amount of elastin and collagen in the lung parenchyma
- ✓ ☒ B) A decrease in the amount of surface tension
- C) Removing the surfactant from alveolar fluid layer
- D) An increase in lung elastic forces

16- An increase in amount of which of the following factors increases the affinity of hemoglobin for oxygen?

- A) Body temperature
- ✓ ☒ B) pH
- C) CO_2 pressure
- D) BPG concentration

17- Which one of the following is NOT a part of respiratory membrane?

- A) A fluid layer of water and surfactants covering the luminal surface of alveolar cells
- B) Basement membranes of endothelial and alveolar cells
- C) Alveolar epithelial cell type-1
- ✓ ☒ D) Smooth muscle cells surrounding the endothelial cells

18- If the diffusing capacity of O₂ is 12 ml/min/mmHg in a patient, how much will be the CO₂ diffusing capacity?

- A) 12 ml/min/mmHg
- B) 120 ml/min/mmHg
- ✓ C) 240 ml/min/mmHg
- D) 360 ml/min/mmHg

19- Which of the following occur during normal inspiration?

- A) Phrenic nerves are inhibited, pleural pressure decreases, alveolar pressure increases
- ✓ B) Phrenic nerves are activated, pleural pressure decreases, alveolar pressure decreases
- C) Phrenic nerves are activated, pleural pressure increases, alveolar pressure increases
- D) Diaphragm contracts, pleural pressure increases, alveolar pressure decreases

20- Which one the following is CORRECT about pulmonary surfactant?

- ✓ A) Thyroid hormone leads to an increase surfactant
- B) Is produced in type I alveolar cells
- C) Most of its composition is related to proteins
- D) Is increased by insulin

21- Which one the following factor has no effect on distribution of pulmonary blood flow in different zones of the lung?

- A) Gravity
- B) Exercise
- ✓ C) Surface tension
- D) Alveolar pressure

22- Which respiratory center provides action potential for contraction of accessory inspiratory muscles during deep inspiration?

- A) DRG
- ✓ B) VRG
- C) Pneumotaxic center
- D) Chemosensitive area

Anatomy: Dr Hasanzadeh

23- Which one is located in the vestibule of nasal cavity?

- ☒ A- Superior nasal concha
- ☐ B- Vibrissae
- ☐ C- Nasolacrimal duct
- ☐ D- Bulla ethmoidalis

24- Which muscle is innervated by recurrent laryngeal nerve?

- ☒ A- Transverse arytenoid
- ☐ B- Inferior constrictor of the pharynx
- ☐ C- Cricothyroid
- ☐ D- Stylopharyngeus

25- Where is the location of Galen anastomosis?

- ☐ A- Rima glottides
- ☐ B- Ventricle
- ☐ C- Saccul
- ☒ D- Piriform recess

26- Superior free border of the conus elasticus makes which one?

- ☐ A- Thyroepiglottic ligament
- ☐ B- Vestibular ligament
- ☒ C- Vocal ligament
- ☐ D- Cricothyroid ligament

27- Which nerve innervates mucosa of the larynx inferior to vocal folds?

- ☒ A- Recurrent laryngeal n.
- ☐ B- Inferior laryngeal n.
- ☐ C- External laryngeal n.
- ☐ D- Internal laryngeal n.

28- In the midclavicular line inferior border of the lung is at the level of which rib?

- ☒ A- 6th
- ☐ B- 8th
- ☐ C- 10th
- ☐ D- 12th

29- Which one is a branch of the maxillary artery?

- A- Anterior ethmoidal
- ☒ B- Ascending palatine
- C- Sphenopalatine
- D- Superior labial

30- Right bronchiolar artery is a branch of which one?

- ☒ A- 3th right intercostal artery
- B- Thoracic aorta
- C- 6th right intercostal artery
- D- Pulmonary artery

31- Where is location of the suprapleural (Sibson's) membrane?

- A- Between first rib and clavicle
- ☒ B- Between first rib and transverse process of the C7
- C- On the medial aspect of the apex
- D- On the anterior aspect of the apex

32- Which one is correct?

- ☒ A- Left main bronchus is longer than right main bronchus
- B- Left main bronchus takes more vertical course
- C- Left main bronchus is wider than right main bronchus
- D- Left main bronchus has 3 lobar bronchi

33- Which part of the pleura is innervated by phrenic nerve?

- ☒ A- Mediastinal pleura
- B- Visceral pleura
- C- Costal pleura
- D- Copula

Histology and Embryology: Dr Rastegar

34- Which of these structures NOT difference between bronchi and bronchioles?

- ☒ A- Epithelial tissue
- B- Cartilage
- C- Glands
- D- Club cells

35- Which type of these cells are NOT in microscopic structure of the respiratory epithelium?

- A- Kolchitsky
- B- Goblet
- C- Brush
- ✓ ☒ D- Supporting

36- In the histological structure of the trachea.....

- A- All epithelial cells have basophilic granules.
- B- The C-shaped cartilage covered by periosteum.
- ✓ ☒ C- Seromucous glands in the lamina propria produce mucus
- D- Hyaline C-shaped cartilage is outermost layer.

37- Where are the Bowman's glands found?

- A- submucosal layer of Trachea
- B- Lamina propria of broncos
- ✓ ☒ C- Olfactory epithelium
- D- Respiratory epithelium of larynx

38- What structures are present in the blood-air barrier?

- ✓ ☒ A- Pneumococcus Type I
- B- Pneumococcus Type II
- C- Pneumococcus Type III
- D- Dust cells

39- Which one of the following statements is TRUE about the Terminal Sac period?

- A- The bronchi will start forming at this period.
- B- Prematurely born fetus are unable to survive.
- C- The epithelium of the terminal sacs will become very thick.
- ✓ ☒ D- Capillaries begin to bulge at this period.

40- Which of the following lining the laryngotracheal diverticulum gives rise to the Glands & muscles of respiratory system, respectively?

- ✓ ☒ A- The endoderm - The Mesoderm
- B- The Mesoderm- Laryngeal clefts
- C- The endoderm - Laryngeal pouches
- D- The mesoderm of 4th and 6th pair of pharyngeal arches